NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
01.	Read each statement carefully and choose the correct answer from the options provided. I. What is Windows Server mainly designed to do? A) Run games and multimedia B) Support enterprise-level management and services C) Edit documents only D) Replace desktop operating systems II. Which Windows Server feature allows management of users and computers in a network? A) Active Directory B) Media Player C) Paint application D) Office suite III. Which Windows Server component allows hosting of websites? A) IIS (Internet Information Services) B) Word processor C) Excel D) Games IV. Which of the following is a primary function of Windows Server? A) Centralized network management B) Playing music C) Drawing and painting D) Running desktop applications only	Answer: I → B II → A III → A IV → A
02.	Read each statement carefully and choose TRUE or FALSE as appropriate. I. Windows Server is designed to support enterprise-level management, data storage, applications, and communications. II. Hyper-V is a Microsoft virtualization technology that allows the creation and management of virtual machines on a physical server.	Answer: I. True II. True III. True IV. False V. False

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	III. Server Core reduces security risks by installing only essential components of Windows Server. IV. A standalone operating system is mainly designed to manage resources and services for multiple network users. V. Network Operating Systems provide advanced features such as centralized management, user authentication, and access control.	
03.	Fill in the blank spaces with the most appropriate word. (Server, 25, DHCP, Router, FTP) A. The powerful machine or application that provides services or resources to clients is called a ..... B. Windows Server Essentials is designed for small businesses with up to ..... users. C. The service that automatically assigns IP addresses to clients in a network is called ..... D. A network device that connects multiple computers in a LAN is called a ..... E. The protocol used to securely transfer files over the Internet is called	Answer: A. Server B. 25 C. DHCP D. Router E. FTP
04.	Read each stem carefully and choose the correct answer from the options provided. I. Which statement best describes a client? A) A device that stores data only B) A device that provides services to others C) A device or application that requests services from a server D) A virtualization platform II. In a network, what is the main role of a client device? A) To request and use resources provided by a server B) To host websites for other devices C) To provide files and services to other clients D) To act as a backup server	Answer: I → C II → A III → A IV → A

<table>
<tr><th>NO</th><th>QUESTIONS (WINDOWS SERVER ADMINISTRATION)</th><th>Answers</th></tr>
<tr><td></td><td>III. Which example correctly represents a client in a network environment?
A) A student's computer accessing the school file server
B) The server storing all user accounts
C) The database hosting company records
D) The network switch

IV. Why is it important to understand the difference between a client and a server?
A) To correctly design and manage network services
B) To increase monitor resolution
C) To install games
D) To replace hardware frequently</td><td></td></tr>
<tr><td>05.</td><td>Match each Windows Server edition in Column A with the correct description in Column B.
<table>
<tr><th>Column A (Edition)</th><th>Column B (Description)</th></tr>
<tr><td>1. Essentials</td><td>A. Free edition focused on virtualization only</td></tr>
<tr><td>2. Standard</td><td>B. Small business, simplified management</td></tr>
<tr><td>3. Datacenter</td><td>C. Supports two OSEs (Operating System Environments)</td></tr>
<tr><td>4. Foundation</td><td>D. Unlimited virtualization rights</td></tr>
<tr><td>5. Hyper-V Server</td><td>E. Entry-level server, supports up to 15 users</td></tr>
</table></td><td>Answer

1 → B
2 → C
3 → D
4 → E
5 → A</td></tr>
</table>

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
06.	Read each statement carefully and choose the correct answer from the options provided. I. Which type of hypervisor runs directly on hardware? A) Hosted B) Type 2 C) Bare-metal D) Desktop II. What is the main advantage of a bare-metal hypervisor over a hosted hypervisor? A) It provides better performance and direct access to hardware B) It can only run on desktops C) It requires a separate operating system to run D) It is mainly used for playing games III. Which hypervisor type relies on an existing operating system to manage virtual machines? A) Bare-metal B) Type 2 (Hosted) C) Type 1 D) None of the above IV. Why is it important to know the difference between Type 1 and Type 2 hypervisors when designing a virtual environment? A) To choose the right hypervisor for performance, security, and management B) To install more software applications C) To increase monitor brightness D) To replace all physical computers	Answer: I → C II → A III → B IV → A
07.	Arrange the steps for creating a bootable USB using Rufus in the correct order: A. Click Start B. Select USB device	Answer: D → B → C → A

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	C. Choose ISO file	

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	C. Choose ISO file D. Download and launch Rufus	
08.	An organization requires centralized file sharing, centralized user authentication, and secure remote access for its staff. Analyse the organization's needs and recommend the most suitable type of operating system to use. Support your answer with clear reasons.	The organization should use a **Network Operating System (Network OS).** Reasoning explained step by step: **First, centralized file sharing:** A Network OS allows files to be stored on a central server, enabling all authorized users to access and share files easily across the network. **Second, user authentication:** A Network *OS* supports centralized user accounts where usernames and passwords are managed from one location, improving security and access control. **Third, remote access:** A Network OS allows administrators and users to access the system remotely using secure network services for management and maintenance. **Why not a Standalone Operating System:** A standalone OS is designed for a single computer only and does not efficiently support centralized management or multiple users in a network environment.
09.	Read each statement carefully and choose the correct answer from the options provided. I. Active Directory is best described as: A) A desktop operating system B) A directory service developed by Microsoft for Windows domain networks	Answer: I → B II → A III → A IV → A

READING SESSION

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	C) A virtualization software D) A network cabling standard II. Which of the following is the primary function of Active Directory? A) Managing users, computers, and resources in a Windows domain B) Playing multimedia files C) Hosting websites only D) Connecting network cables III. Active Directory is mainly used in which type of network? A) Windows domain networks B) Home offline networks only C) Gaming networks D) Standalone PCs without networking IV. Who developed Active Directory? A) Microsoft B) Linux Foundation C) Cisco D) Apple	
10.	Fill in the blank spaces with the most appropriate word or phrase. (Domain Controller, DNS Server, Web Server, File Server, Mail Server) A. A server that runs Active Directory Domain Services is known as a ..... B. A server that translates domain names to IP addresses is called a ..... C. A server that hosts websites and web applications is called a ..... D. A server that stores and manages files for network users is called a ..... E. A server that handles sending and receiving emails is called a	Answer: A. Domain Controller B. DNS Server C. Web Server D. File Server E. Mail Server
11.	Read each statement and indicate whether it is TRUE or FALSE. i. Active Directory provides authentication, authorization, and centralized management of network resources. ii. Active Directory Domain Services (AD DS) stores and manages directory information.	Answer: i. TRUE ii. TRUE iii. TRUE

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NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	iii. DNS is required for Active Directory to function properly. iv. A Domain Controller hosts AD DS. v. Group Policy allows administrators to centrally manage user and computer settings	iv. TRUE v. TRUE
12	Read each stem carefully and choose the correct answer from the options provided. I. Which Active Directory component is used to logically organize users and computers? A) Forest B) Tree C) Organizational Unit (OU) D) Schema II. Which component represents a collection of multiple domains in Active Directory? A) OU B) Forest C) Schema D) Tree III. In Active Directory, which component is a collection of domains that share a common namespace? A) Forest B) Tree C) OU D) Schema IV. Which Active Directory component defines the structure and attributes of objects stored in the directory? A) Schema B) OU C) Tree D) Forest	Answer: I → C II → B III → B IV → A

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NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
13	Fill in the blanks with the most appropriate word or phrase. (Global Catalog, Domain Controller, Schema, Forest, Read-Only Domain Controller) a) The ……..contains a partial copy of every object in the Active Directory …….. b) It is hosted on a ……..and helps users locate resources across the …….. c) A ……..may also store directory information with limited write permissions.	Answer: a) Global Catalog and Forest b) Domain Controller and Forest c) Read-Only Domain Controller
14	Read each statement carefully and choose the correct answer from the options provided. I. Why are Organizational Units (OUs) important in Active Directory management? A) They install server hardware B) They help organize objects and delegate administrative control C) They manage internet connectivity D) They replace domain controllers II. Which of the following is something OUs cannot do in Active Directory? A) Organize users and computers B) Apply Group Policies C) Delegate administrative control D) Replace domain controllers III. Which statement about OUs is true regarding administrative delegation? A) OUs install software automatically B) OUs provide internet connectivity management C) OUs automatically upgrade servers D) OUs allow delegation of control without giving full domain rights	Answer: I → B II → D III → D IV → D

<table>
<tr><th>NO</th><th>QUESTIONS (WINDOWS SERVER ADMINISTRATION)</th><th>Answers</th></tr>
<tr><td></td><td>IV. Which of the following is NOT a primary function of OUs in Active Directory?
A) Contain users, groups, and computers
B) Help apply Group Policies
C) Facilitate management of network objects
D) Replace domain controllers</td><td></td></tr>
<tr><td>15</td><td>Match the Active Directory concepts in Column A with their correct descriptions in Column B.

<table><tr><th>Column A</th><th>Column B</th></tr><tr><td>1. Forest</td><td>A. A container used to organize objects within a domain</td></tr><tr><td>2. Global Catalog</td><td>B. A logical grouping of computers, users, and resources</td></tr><tr><td>3. Trust Relationship</td><td>C. A partial replica used for cross-domain searches</td></tr><tr><td>4. Domain</td><td>D. A collection of domains sharing a common schema</td></tr><tr><td>5. Organizational Unit (OU)</td><td>E. A security link allowing access between domains</td></tr></table></td><td>Answer

1 → D
2 → C
3 → E
4 → B
5 → A</td></tr>
<tr><td>16</td><td>Read each stem carefully and choose the correct answer from the options provided.

I. Which Active Directory object represents an individual authorized to access network resources?
A) Computer account
B) Security group
C) User account
D) Organizational Unit</td><td>Answer:

I → C
II → B
III → D
IV → D</td></tr>
</table>

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	II. Which AD object can be used to group multiple users for easier permission management? A) User account B) Security group C) Computer account D) Organizational Unit III. Which AD object can contain users, groups, and computers to simplify administration? A) Security group B) User account C) Computer account D) Organizational Unit IV. Which AD object is mainly used to assign permissions and policies to multiple objects at once? A) User account B) Security group C) Computer account D) Organizational Unit	
17	Read each statement carefully and choose the correct answer from the options provided. I. An organization wants to manage users and computers centrally with authentication and security policies. Which service should be installed? A) Local Group Policy B) Standalone Operating System C) Active Directory Domain Services D) DNS only II. A company needs to control user logins, passwords, and permissions across multiple computers. Which service should they use? A) Active Directory Domain Services B) DNS server only	Answer: I → C II → A III → C IV → D

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NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	C) Individual desktops D) Local Group Policy III. To apply security policies and manage resources across all employees in a domain, which Windows Server role is required? A) Desktop application B) Media services C) Active Directory Domain Services D) Standalone Operating System IV. A business wants to ensure that only authorized users can access network resources and that policies are consistent for everyone. Which service should be implemented? A) Standalone Operating System B) Local software installation C) DNS server D) Active Directory Domain Services	
18	Arrange the steps for joining a client computer to a domain in the correct order. A. Restart the computer B. Enter domain credentials C. Open System Properties D. Select "Domain" and enter the domain name	Answer: C → D → B → A each
19	Consider the DHCP Discover phase. Indicate whether the following statements are true or false. A) DHCP Discover messages are broadcast to locate available DHCP servers. B) The client's MAC address is used to target the broadcast. C) DHCP Discover messages are sent after the client receives an IP address.	Answer: A. True B. True C. False D. True

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	D) The destination MAC address in the broadcast is FF:FF:FF:FF:FF:FF.	
20	Fill in the blanks with the correct terms. (IP address, renew, release, DHCP server, lease time) a) The DHCP lease time defines how long a client can use an assigned …… b) After expiration, the client must …… the IP address from the …… c) If the client no longer needs the address, it may …… it before the …… ends.	Answer: a) IP address b) Renew, DHCP server /2marks c) Release, lease time /2marks
21	Read each stem carefully and choose the correct answer from the options provided. I. Which of the following command pairs should the administrator use to release and renew an IP address? A) ipconfig /flush and ping /release B) netstat /renew and ipconfig /flush C) ping /renew and ipconfig /release D) ipconfig /renew and ipconfig /release II. An administrator wants to free the current IP and request a new one on a Windows machine. Which commands are correct? A) ipconfig /renew and ipconfig /release B) ping /renew and ipconfig /release C) netstat /renew and ipconfig /flush D) ipconfig /refresh and ping /release III. To troubleshoot a network connectivity issue by resetting the IP address, which command pair should be executed? A) ipconfig /refresh and ping /release B) ipconfig /renew and ipconfig /release	Answer: I → D II → A III → B IV → C

READING SESSION

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	C) ipconfig /release and ipconfig /renew D) ping /renew and ipconfig /release IV. Which command pair correctly allows releasing the current IP and acquiring a new IP from the DHCP server? A) netstat /renew and ipconfig /flush B) ipconfig /flush and ping /release C) ping /renew and ipconfig /release D) ipconfig /release and ipconfig /renew	
22	Fill in the blanks with the most appropriate word or phrase. (Local Server, Network Connections, Ethernet, Internet Protocol Version 4 (IPv4), Properties) In Windows GUI, DHCP client settings can be configured by accessing Server Manager → a)…… →b) …… → c)…… →d) …… → e)……, where you can select DHCP client or static IP assignment.	Answer: a) Local Server b) Network Connections c) Ethernet d) Internet Protocol Version 4 (IPv4) e) Properties
23	A client computer sends a DHCP Discover message and receives DHCP Offer messages from more than one DHCP server. Explain how the client responds to multiple DHCP Offers and why this process is important in a network environment.	Answer: 1. Client response to multiple DHCP Offers • The client evaluates all received DHCP Offers. • It selects one offer, usually the first offer or based on preferred criteria (e.g., lease time, server reliability). • The client then sends a DHCP Request message to the chosen DHCP server to accept the IP address. **2. Importance of this process in a network environment** • Ensures that the client obtains a valid IP address without conflicts.

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		conflicts.

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
		• Allows multiple DHCP servers to coexist, providing redundancy and high availability in the network. • Helps maintain proper IP address allocation and avoids duplication, supporting stable network operations.
24	Read each stem carefully and choose the correct answer from the options provided. I. Which DNS record is used to map a domain name to an IPv4 address? A) A B) AAAA C) CNAME D) MX II. Which DNS record is used to map a domain name to an IPv6 address? A) CNAME B) AAAA C) A D) MX III. Which DNS record is used to alias one domain name to another? A) MX B) AAAA C) A D) CNAME IV. Which DNS record specifies the mail server for a domain? A) A B) MX C) AAAA D) CNAME	Answer: I → A II → B III → D IV → C

<table>
<tr><th>NO</th><th>QUESTIONS (WINDOWS SERVER ADMINISTRATION)</th><th>Answers</th></tr>
<tr><td>25</td><td>Match Column A with the correct description in Column B.
Write the letter of the correct answer from Column B next to each item in Column A.

<table><tr><th>Column A</th><th>Column B</th></tr><tr><td>A. DNS A Record</td><td>1. Translates a domain name into an IPv4 address</td></tr><tr><td>B. Domain Name</td><td>2. The numeric address used to identify a device on a network</td></tr><tr><td>C. DNS Server</td><td>3. A system that stores and resolves domain name records</td></tr><tr><td>D. IPv4 Address</td><td>4. A human-readable name used to access network resources</td></tr><tr><td>E. DNS Resolution</td><td>5. The process of converting a domain name into an IP address</td></tr></table></td><td>Answer:

A → 1
B → 4
C → 3
D → 2
E → 5</td></tr>
<tr><td>26</td><td>Match Column A with the correct action in Column B that is performed when updating DNS client configuration on a Windows computer.

<table><tr><th>Column A</th><th>Column B</th></tr><tr><td>A. Open Network Settings</td><td>1. Select Internet Protocol Version 4 (IPv4) and click Properties</td></tr><tr><td>B. Change Adapter Options</td><td>2. Access the list of available network connections</td></tr><tr><td>C. Network Adapter Properties</td><td>3. Enter preferred and alternate DNS server addresses</td></tr></table></td><td>Answer:
A → 4
B → 2
C → 1
D → 3
E → 5</td></tr>
</table>

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)		Answers
	D. IPv4 Properties	4. Open Settings → Network & Internet	
	E. DNS Server Configuration	5. Apply and save the new DNS configuration	
27	Explain why a company should use public DNS services, such as Google DNS, instead of relying solely on its ISP's DNS servers. In your answer, discuss how public DNS affects reliability, performance, and geographic distribution, and justify why it is the better choice. Give examples to support your explanation.		**Answer:** **Reliability** • Public DNS uses redundant servers and failover mechanisms, ensuring minimal downtime if some servers fail. **Performance** • Public DNS reduces lookup delays and improves response times due to optimized and distributed infrastructure. **Geographic Distribution** • Public DNS servers are located in **multiple regions**, serving users from nearby locations efficiently and reducing latency. **Justification / Examples** • Example: A multinational company using Google DNS experiences faster website access, fewer downtime incidents, and consistent performance globally. • Reasoning: Public DNS provides fault tolerance, scalability, and global coverage, which a single ISP DNS cannot guarantee.

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
28	You are assigned to design and implement DNS for a startup company. Explain: a) The DNS server you would install and why. b) The essential DNS records to configure. c) If you would configure external DNS (like Google or Cloudflare), and why. d) How you would confirm that domain names correctly resolve to IP addresses.	**Answer:** **a) DNS Server Installation** I would install Windows Server DNS in a Windows-based environment or BIND if using Linux. These DNS servers are reliable, widely supported, and allow full management of DNS zones and records. **b) Essential DNS Records** **A Record**: Maps a domain name to an IPv4 address. **CNAME Record**: Links an alias name to the main (canonical) domain. **MX Record**: Routes email traffic for the domain. **NS Record**: Specifies the authoritative name servers for the domain. **c) External / Public DNS Configuration** I would optionally configure public DNS services such as **Google DNS** or **Cloudflare DNS** to enhance performance and availability. **Redundancy**: Ensures DNS resolution if the internal DNS server fails. **Speed**: Improves resolution through global caching and optimized infrastructure. **Reliability and Security**: Protects against DNS attacks and reduces downtime. **d) Verification of DNS Resolution** Use **ping** to verify that the domain resolves to an IP address. Use **nslookup** or **dig** to check DNS records, TTL values, and authoritative servers. Optionally, use online tools such as **MXToolbox** or **DNSChecker** to confirm global DNS propagation.

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
29	Read each stem carefully and choose the correct answer from the options provided. I. Which of the following web servers is specifically optimized for Windows and ASP.NET? A) Apache B) Nginx C) IIS D) LiteSpeed II. Which web server is known for high performance and handling large concurrent connections efficiently? A) IIS B) Nginx C) Lighttpd D) Tomcat III. Which web server is open-source and widely used on Linux environments? A) Apache B) Microsoft IIS C) LiteSpeed D) Nginx IV. Which web server is lightweight and often used for shared hosting environments? A) Tomcat B) Apache C) Nginx D) LiteSpeed	Answer: I → C II → B III → A IV → D
30	Read each statement carefully and write TRUE if the statement is correct or FALSE if it is incorrect	Answer:

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NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	a) To host a virtual website using host headers, a new website must be created in IIS Manager. b) Host headers allow multiple websites to share the same IP address and port number. c) A host header is configured by entering the website's domain name in the bindings settings. d) DNS configuration is not required when using host headers in IIS e) Testing a host-header website can be done by browsing the domain name after configuration.	a) TRUE b) TRUE c) TRUE d) FALSE e) TRUE
31	Analyse the difference between a real host and a virtual host. In your explanation, compare how each one uses server resources and how websites are hosted on them.	**Answer:** 1. **Real Host – 5 marks** A real host is a dedicated physical server where all hardware resources (CPU, memory, storage, network) are used by a single organization or website. This provides better performance, higher security, and full control over server settings. 2. **Virtual Host – 5 marks** A virtual host allows multiple websites or users to share the same physical server. Resources are divided logically so each site behaves as if it has its own server. This is cost-effective and efficient, but performance and control may be limited compared to a real host.
32	A startup plans to host multiple websites on a single server. I. Recommend a suitable web server software for this purpose. II. Justify your recommendation by explaining how the chosen web server performs in terms of: a) Performance	**Answer:** **I. Recommended web server** **Recommended web server: Apache HTTP Server**

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	b) Security c) Ease of management	**II. Justification (8 marks)** a) **Performance** Apache efficiently handles multiple websites using virtual hosts and provides stable performance for small and medium traffic levels. It also supports caching and compression to improve speed. b) **Security** Apache offers strong security features such as SSL/TLS support, access control, authentication modules, and regular security updates, which help protect hosted websites. c) **Ease of management** Apache is easy to configure, well-documented, and widely supported. Many administrators are familiar with it, making setup and maintenance simple.
33	Design a complete deployment plan for a web application on Windows Server, including server selection, web server type, security configuration, and testing methods.	**Answer:** **1. Server Selection** o Choose an appropriate Windows Server edition (e.g., 2019 or 2022). o Ensure hardware meets requirements: CPU, RAM, storage, network. o Consider virtualization for dev/test/prod environments. **2. Web Server Type (3.5 marks)** o Install and configure IIS (Internet Information Services). o Enable required features/modules based on application (.NET, PHP, etc.). o Configure application pools for performance and isolation. **3. Security Configuration**

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
		○ Enable Windows Firewall and allow only HTTP/HTTPS traffic. ○ Configure SSL/TLS certificates for secure communication. ○ Set NTFS permissions properly on application folders. ○ Apply updates/patches and antivirus protection. ○ Optionally implement IP restrictions or URL authorization. **4.** Testing Methods (3.5 marks) ○ Conduct functional testing to ensure features work correctly. ○ Perform load/performance testing *for* expected traffic. ○ Conduct security testing (vulnerability scanning, penetration testing). ○ Test backup/recovery and disaster recovery processes.
34	Read each stem carefully and choose the correct answer from the options provided. I. Which of the following is the correct meaning of the acronym FTP used in networking for transferring files between computers? A) File Transport Protocol B) File Transfer Protocol C) Fast Transfer Program D) File Transmission Process II. Which protocol is commonly used to upload and download files from a server? A) FTP B) HTTP C) SMTP D) DNS III. Which protocol allows secure file transfer using encryption? A) SFTP	Answer: I → B II → A III → C IV → D

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	B) FTP C) HTTP D) POP3 IV. Which protocol is mainly used for sending emails over a network? A) FTP B) DNS C) HTTP D) SMTP	
35	Write TRUE or FALSE for each of the following statements: a) LiteSpeed web server is preferred for high-traffic websites because it can handle more concurrent connections efficiently. b) LiteSpeed web server is slower than Apache when serving static content. c) LiteSpeed has built-in caching features that improve website performance. d) LiteSpeed cannot handle HTTPS traffic efficiently.	Answer: a) TRUE b) FALSE c) TRUE d) FALSE
36	When hosting a virtual website on IIS using host headers, which of the following actions are essential? (Select two correct answers) a) Configure the website with a unique host header (domain name) in IIS b) Assign the same port and IP address for multiple websites without host headers c) Install SSL certificates on the server before creating any websites d) Bind the website to the appropriate IP address, port, and host header	Answer: a) Configure the website with a unique host header (domain name) /2marks d) Bind the website to the appropriate IP address, port, and host header
37	A company needs to transfer files in different scenarios: managing website files, simple internal transfers on a Windows network, enterprise-level secure transfers, and supporting high availability with multiple protocols.	**Answer:** **1. FileZilla** • Correct use case: cross-platform, user-friendly FTP tasks

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	Evaluate the FTP clients FileZilla, Cerberus FTP Server, SolarWinds Serv-U MFT, and Complete FTP, and explain which client is most appropriate for each scenario. Justify your choices based on usability, security, and features.	• Justification: simplicity and ease of use **2. Cerberus FTP Server** • Correct use case: internal Windows networks, easy setup • Justification: simple file transfers **3. SolarWinds Serv-U MFT –** • Correct use case: large enterprises, secure transfers, compliance • Justification: detailed user management **4. Complete FTP –** • Correct use case: organizations needing high availability and multiple protocol support • Justification: advanced features
38	You are a network administrator for a small web company. Explain how you would set up a secure FTP system for managing client website files. In your answer, i. Describe the server software you would use and why, • how you would organize folders and set permissions for admins and clients, ii. the client software you would choose and why, • which FTP protocol you would use and why it is secure, and • how you would test that the system works properly and securely.	**Answer:** **i. Server Software Choice –** • Correct choice of FileZilla Server • Justification: easy setup, secure protocols, multi-client management **. Root Folder and Permissions – 4 marks** • Organizing root folder for clients • Admin permissions: Read, Write, Execute • Client permissions: restricted to their own files **ii. Client Software Choice**

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
		• Correct choice of FileZilla Client • Justification: multi-platform, secure uploads/downloads **. FTP Protocol –** • Correct choice of SFTP • Justification: encrypts data, protects files **. Testing and Validation** •Test connection for admin and clients •Verify permissions and file access
39	Read each statement carefully and write TRUE or FALSE. 1. Load balancing improves performance and reliability by distributing traffic among servers. 2. DNS-based load balancing encrypts traffic between clients and servers. 3. Hardware load balancers operate only at the network layer (Layer 3). 4. Virtual load balancers combine features of hardware and software solutions. 5. Capacity planning helps prepare for future traffic demands.	Answer: 1 → True 2 → False 3 → False 4 → True 5 → True

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
40	Complete the sentences using the most appropriate term. (Hardware Load Balancer, Server Pool, Software Load Balancer, DNS-based Load Balancing, Load Balancing) 1. …… is a group of servers used to handle incoming requests. 2. Traffic distribution methods like …… ensure even allocation across servers. 3. …… is a physical or virtual device that manages traffic distribution across servers. 4. A …… runs as an application on a standard server or cloud to manage traffic. 5. …… uses the Domain Name System to distribute traffic among multiple servers.	Answer: 1. Server Pool 2. Load Balancing 3. Hardware Load Balancer 4. Software Load Balancer 5. DNS-based Load Balancing
41	A company is designing its IT infrastructure to achieve high system reliability, improved performance, and efficient management of network traffic. Explain how the following concepts can be applied in the company's server and network environment: • High Availability • Performance Optimization • DNS-based Load Balancing • Disaster Recovery • Server Pool In your answer, clearly show the purpose of each concept and how it would be implemented in a real organization.	**Answer:** **High Availability** ensures that services remain accessible even when a system component fails. In an organization, this can be implemented using redundant servers or failover clustering so another server takes over automatically. **Performance Optimization** improves system speed and responsiveness. A company can optimize performance by upgrading hardware, tuning server settings, and managing network resources efficiently. **DNS-based Load Balancing** distributes user requests across multiple servers using DNS records. For example, a website domain can resolve to different IP addresses to share traffic among web servers.

READING SESSION

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
		Disaster Recovery focuses on restoring systems and data after major failures. This can be achieved through regular backups, off-site storage, and a documented recovery plan. **Server Pool** refers to multiple servers working together to handle workloads. An organization can group application servers into a pool so requests are served by any available server.
42	A growing company wants to ensure its IT systems are reliable, can handle increasing traffic, and recover quickly from failures. Evaluate the strategies or tools the company could implement to achieve these goals. In your answer, identify the most suitable approach for distributing traffic, handling high loads, planning capacity, and ensuring system recovery, and justify each choice.	**Answer:** **DNS-based Load Balancing** • **Identification :** Distributes client requests across multiple servers using DNS. • **Justification :** Balances traffic globally, ensuring no single server is overloaded. **Hardware Load Balancer** • **Identification :** A physical device managing traffic at the application layer. • **Justification :** Provides high-performance traffic management for critical applications. **Software Load Balancer** • **Identification :** An application running on a standard server or cloud to manage traffic. • **Justification :** Flexible and cost-effective for smaller or cloud-based environments. **Capacity Planning**

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READING SESSION

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
		• **Identification :** Planning resources to handle future traffic and avoid overload. • **Justification :** Ensures IT systems scale properly with growth, preventing downtime. **Disaster Recovery** • **Identification :** Plans to restore IT systems after failures. • **Justification :** Guarantees business continuity by minimizing downtime after a system failure.
43	A new e-commerce company expects heavy traffic on its website. You have been tasked with designing a load balancing solution. In your answer, describe: i. The type of load balancer you would implement (hardware, software, or virtual) and justify your choice. ii. How the server pool will be structured to handle requests efficiently. iii. The method you will use to distribute traffic among servers. iv. How you will plan for capacity to accommodate peak traffic. v. Measures for disaster recovery to ensure high availability.	**Answer:** **i. Load Balancer :** Use a virtual load balancer – flexible, scalable, cost-effective, and supports cloud or on-premises deployment. **ii. Server Pool Structure :** Multiple servers grouped by function: web servers, application servers, database servers. Each layer has 2–3 servers for redundancy and fault tolerance. **iii. distribution :** Use round-robin or least-connection algorithms. Optionally, use DNS-based balancing for global traffic. **iv. Capacity Planning :** Estimate peak traffic, add extra servers or cloud instances, monitor performance, and use auto-scaling to handle spikes. **v. Disaster Recovery :** Implement redundant load balancers, replicate data, enable failover mechanisms, and test recovery plans regularly.

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READING SESSION

<table>
<tr><th>NO</th><th>QUESTIONS (WINDOWS SERVER ADMINISTRATION)</th><th>Answers</th></tr>
<tr><td>44</td><td>Read each statement carefully and choose the correct answer from the options provided.

I. Which of the following is a routine server maintenance task?
A) Installing games
B) Monitoring server logs
C) Playing media
D) Browsing the internet

II. Which task helps ensure that the server's software is up to date?
A) Updating server applications
B) Playing videos
C) Installing games
D) Browsing websites

III. Which task involves checking hardware performance and identifying potential failures?
A) Gaming
B) Monitoring CPU and memory usage
C) Watching media
D) Using a browser

IV. Which task helps prevent unauthorized access by managing user accounts and permissions?
A) Playing games
B) Monitoring server logs
C) Editing documents
D) Managing user accounts and permissions</td><td>Answer:

I → B
II → A
III → B
IV → D</td></tr>
<tr><td>45</td><td>Match the maintenance task with its purpose.

<table><tr><th>Column A</th><th>Column B</th></tr><tr><td>1. Disk cleanup</td><td>a) Protects data from loss</td></tr></table></td><td>Answer:

1 → b
2 → a
3 → c
4 → d
5 → e</td></tr>
</table>

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	Wrong up	

5 → e

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
	2. Backup — b) Frees space and removes junk files 3. Antivirus update — c) Prevents malware infections 4. Software patch — d) Fixes bugs and security vulnerabilities in applications 5. Defragmentation — e) Organizes fragmented data on the hard drive to improve performance	
46	Fill in the blanks with the correct terms related to server maintenance. Using (Checking resource usage, Regular backups, Security updates, Monitoring logs, Installing software, Disk formatting) Regular application of 1…… ensures that the server runs with the latest security patches. In addition, performing 2……, 3……, and 4…… helps keep the server secure, stable, and running efficiently.	Answer: 1. Security updates / patches (1mark) 2. Monitoring logs(1mark) 3. Checking resource usage(1mark) 4. Regular backups(1mark)
47	You notice a server is running slowly. Which maintenance actions should you perform first? (Select two answers) a) Monitor logs b) Disk cleanup c) Check resource usage d) Reinstall the OS	Answer: a) Monitor logs / c) Check resource usage/
48	A server requires regular maintenance to ensure performance, security, and reliability. Explain the correct sequence of steps you would follow for server maintenance and justify why each step should be performed in that order.	**Answer:** **Backup Critical Data** • Before performing any maintenance, back up important files, databases, and system configurations.

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
		• This ensures that data is safe in case updates or disk cleanups cause issues. **Apply Updates/Patches** • Install operating system updates, application patches, and security updates. • Keeping software up-to-date prevents vulnerabilities and improves stability. **Disk Cleanup** • Remove temporary files, logs, and unused data to free storage and improve performance. • Helps avoid system slowdowns and ensures smooth operation. **Monitor Logs and Performance** •Check system logs for errors or unusual activity and monitor CPU, memory, and network usage. •Ensures the system is running efficiently and identifies any potential problems early.
49	A network administrator observes several issues in the organization's IT infrastructure: server downtime during peak hours, frequent disk errors, and slow network response. For each issue, explain what evaluation actions you would take to identify the cause and propose a solution.	**Answer:** **Server downtime during peak hours** (3marks) • Evaluation Action: Analyze server log files to identify patterns and peak load times. • **Solution:** Optimize server performance, redistribute workload, or upgrade server resources to handle peak demand.

		demand.

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
		Frequent disk errors (3marks) • **Evaluation Action:** Check disk health using diagnostics tools (e.g., SMART, chkdsk). • **Solution:** Replace or repair faulty hardware components, or implement RAID for redundancy. **Slow network response** (4marks) • **Evaluation Action:** Monitor network traffic, check bandwidth usage, and inspect network configurations. • **Solution:** Adjust network settings, upgrade bandwidth, or optimize traffic routing to improve performance.
50	Design a server maintenance plan for a medium-sized company that hosts multiple critical applications. Discuss Your plan should include 1. Backup strategy and frequency 2. Update and patch management 3. Disk cleanup and defragmentation schedule 4. Security monitoring measures 5. Performance monitoring and capacity planning	**Answer:** **1. Backup Strategy and Frequency –** • Daily incremental and weekly full backups • Onsite and offsite/cloud storage for redundancy • Regular testing of backups for data integrity **2. Update and Patch Management –** • Regular scheduling of OS, application, and security patches • Testing patches in a staging environment before deployment • Enabling automatic updates for critical security patches **3. Disk Clean-up and Defragmentation** • Weekly disk cleanup and removal of temporary files/logs • Monthly defragmentation for HDDs or SSD optimization

NO	QUESTIONS (WINDOWS SERVER ADMINISTRATION)	Answers
		4. Security Monitoring Measures – 4 marks • Installing/updating antivirus and anti-malware • Monitoring server logs for suspicious activity • Implementing firewalls and IDS • Enforcing strong access controls **5. Performance Monitoring and Capacity Planning** • Tracking CPU, memory, disk, and network usage • Identifying bottlenecks and scaling resources • Planning for peak loads and future growth